ccd_code

January 13, 2026

[335]:
```
from astropy.io import fits
import numpy as np
from matplotlib import pyplot as plt
from glob import glob
import seaborn as sns
from scipy import stats
import matplotlib.mlab as mlab
from scipy.signal import find_peaks
from scipy.optimize import curve_fit

%matplotlib inline
sns.set()
```

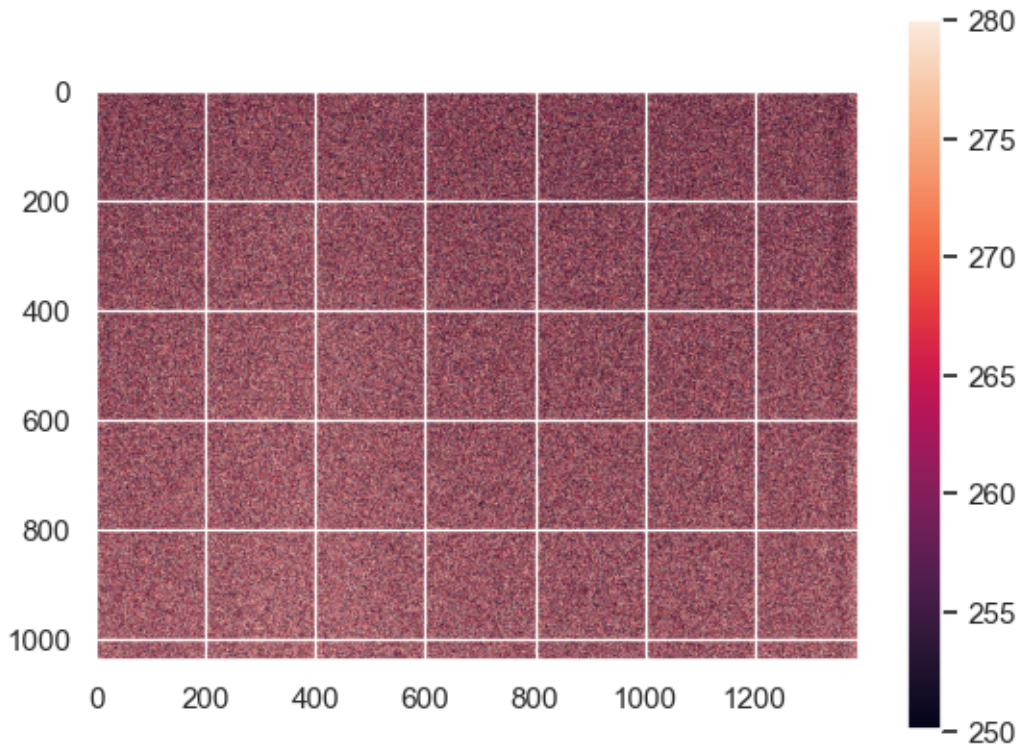
1 bias

[336]:
```
image = fits.open('biases_030bias.fit')
image[0].data
print(np.min(image[0].data), np.max(image[0].data))
```

```
182 352
```

[337]:
```
pixelval = image[0].data
plt.imshow(pixelval, vmin=250 ,vmax=280)
plt.colorbar()
```

[337]:
```
<matplotlib.colorbar.Colorbar at 0x11c0e88d0>
```



[338]:
```
biases = glob('biases*')
print(biases)
```

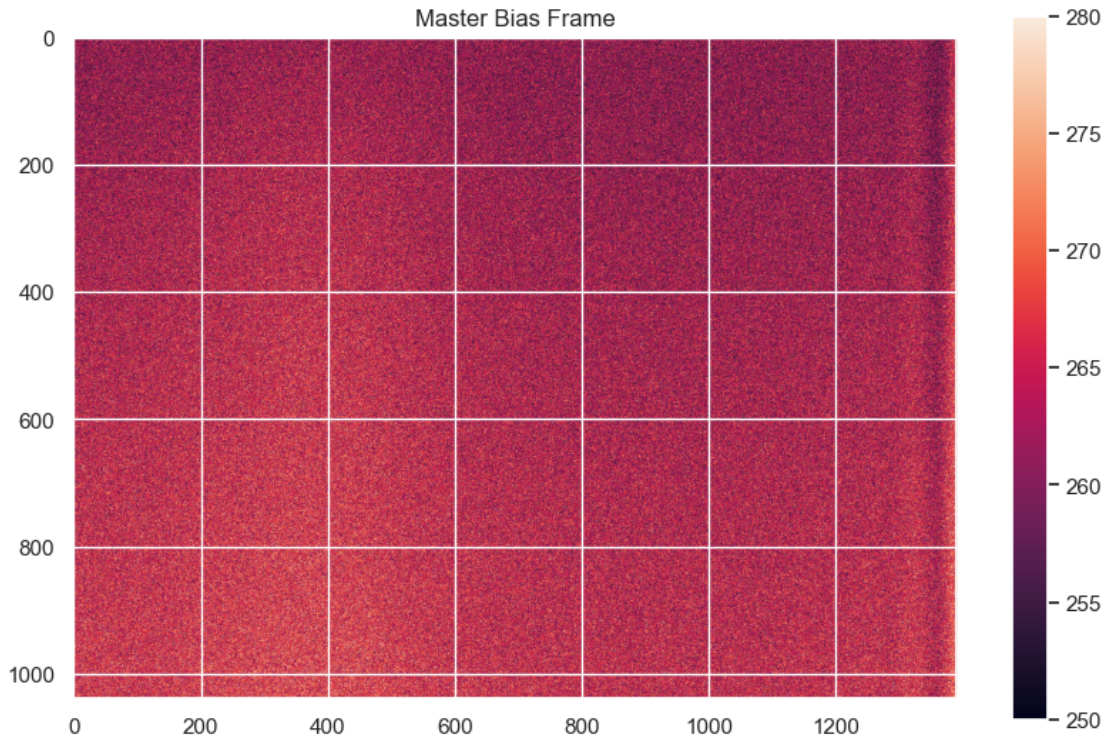
```
['biases_031bias.fit', 'biases_038bias.fit', 'biases_039bias.fit',
'biases_030bias.fit', 'biases_032bias.fit', 'biases_033bias.fit',
'biases_036bias.fit', 'biases_037bias.fit', 'biases_035bias.fit',
'biases_034bias.fit']
```

[339]:
```
bias_data = np.stack([fits.getdata(f).astype(float) for f in biases], axis=0)

master_bias = np.mean(bias_data, axis=0)
```

[386]:
```
plt.figure(figsize=(10,8))
plt.imshow(master_bias, vmin=250, vmax=280)
#plt.scatter(900,500)
plt.colorbar(shrink=0.8)
plt.title('Master Bias Frame')

print(np.mean(master_bias), np.std(master_bias))

master_bias = master_bias / 263.70
print(np.mean(master_bias))
```

```
263.6975490036665 5.5433156610801895
```

```
0.9999907053608893
```



[341]:
```
mean_box = master_bias[800:1000, 400:600].mean()
std_box = master_bias[800:1000, 400:600].std()

print(f'mean: {mean_box}\nstd: {std_box}')
```

```
mean: 266.46279749999997
std: 5.365270796893084
```

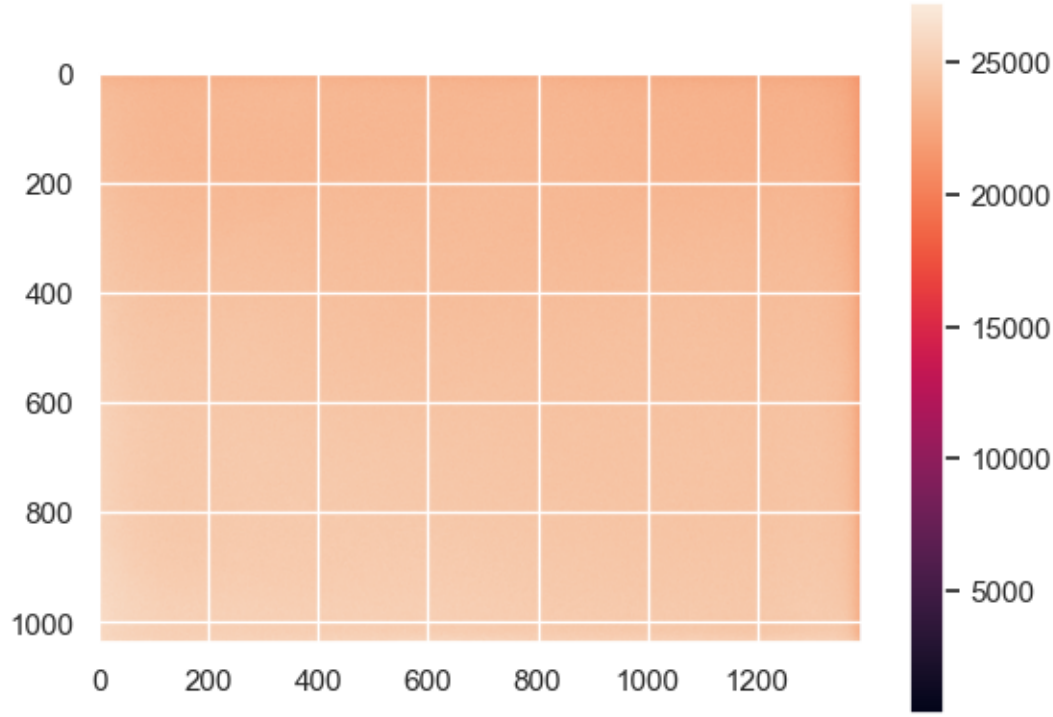
2 flat

[342]:
```
imagef = fits.open('flats_001flat.fit')
imagef[0].data
print(np.min(imagef[0].data), np.max(imagef[0].data))
```

```
320 27111
```

[343]:
```
pixelvalf = imagef[0].data
plt.imshow(pixelvalf, vmin=320 ,vmax=27250)
plt.colorbar()
```

[343]: <matplotlib.colorbar.Colorbar at 0x11bc50a90>



[344]:
```
flats = glob('flats_*')
print(flats)
```

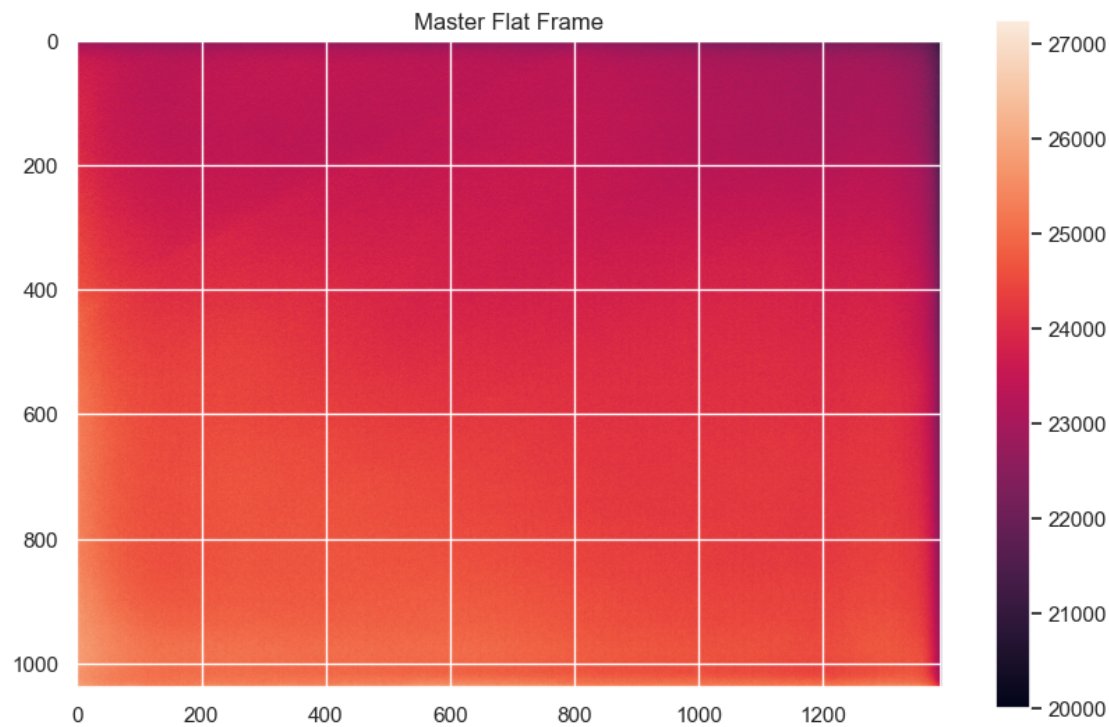
```
['flats_008flat.fit', 'flats_001flat.fit', 'flats_010flat.fit',
'flats_009flat.fit', 'flats_002flat.fit', 'flats_003flat.fit',
'flats_006flat.fit', 'flats_007flat.fit', 'flats_005flat.fit',
'flats_004flat.fit']
```

[345]:
```
flat_data = []
for f in flats:
    dataf = fits.getdata(f).astype(float)
    datafb = dataf - master_bias
    flat_data.append(datafb)

flat_data = np.stack(flat_data, axis=0)
master_flat = np.mean(flat_data, axis=0)
```

[346]:
```
plt.figure(figsize=(10,8))
plt.imshow(master_flat, vmin=20000, vmax=27250)
#plt.scatter(900,500)
plt.colorbar(shrink=0.8)
plt.title('Master Flat Frame')

print(np.mean(master_flat), np.std(master_flat))
```

```
master_flat = master_flat / 24027.38
print(np.mean(master_flat))
```

```
24027.38192069325 907.408904417156
1.0000000799376898
```



[347]:
```
mean_boxf = master_flat[800:1000, 400:600].mean()
std_boxf = master_flat[800:1000, 400:600].std()

print(f'mean: {mean_boxf}\nstd: {std_boxf}')
```

```
mean: 1.0321517471942427
std: 0.007674310620867742
```

3 readnoise and gain

3.1 indiv. readnoise/gain

[348]:
```
print(biases)
print(flats)
```

```
['biases_031bias.fit', 'biases_038bias.fit', 'biases_039bias.fit',
'biases_030bias.fit', 'biases_032bias.fit', 'biases_033bias.fit',
'biases_036bias.fit', 'biases_037bias.fit', 'biases_035bias.fit',
```

```
'biases_034bias.fit']
['flats_008flat.fit', 'flats_001flat.fit', 'flats_010flat.fit',
'flats_009flat.fit', 'flats_002flat.fit', 'flats_003flat.fit',
'flats_006flat.fit', 'flats_007flat.fit', 'flats_005flat.fit',
'flats_004flat.fit']
```

[349]:
```
bias_data = []
for f in biases:
    data_b = fits.getdata(f).astype(float)
    bias_data.append(data_b)

flat_data = []
for f in flats:
    dataf = fits.getdata(f).astype(float)
    datafb = dataf - master_bias
    flat_data.append(datafb)

biases = bias_data
flats = flat_data
```

[350]:
```
biasdif1 = biases[0] - biases[1]
flatdif1 = flats[0] - flats[1]

mean_bias1 = np.mean(biases[0]) + np.mean(biases[1])
mean_flat1 = np.mean(flats[0]) + np.mean(flats[1])

std_bias1 = np.std(biasdif1)
std_flat1 = np.std(flatdif1)

gain1 = (mean_flat1 - mean_bias1) / (std_flat1**2 - std_bias1**2)
readnoise1 = gain1 * (std_bias1) / (np.sqrt(2))

print(f'gain = {gain1}\nreadnoise = {readnoise1}')
```

```
gain = 0.2696555886000731
readnoise = 4.470695042713425
```

[351]:
```
biasdif2 = biases[2] - biases[3]
flatdif2 = flats[2] - flats[3]

mean_bias2 = np.mean(biases[2]) + np.mean(biases[3])
mean_flat2 = np.mean(flats[2]) + np.mean(flats[3])

std_bias2 = np.std(biasdif2)
std_flat2 = np.std(flatdif2)

gain2 = (mean_flat2 - mean_bias2) / (std_flat2**2 - std_bias2**2)
readnoise2 = gain2 * (std_bias2) / (np.sqrt(2))
```

```
print(f'gain = {gain2}\nreadnoise = {readnoise2}')
```

```
gain = 0.26984208637440976
readnoise = 4.471747003084644
```

[352]:
```
biasdif3 = biases[4] - biases[5]
flatdif3 = flats[4] - flats[5]

mean_bias3 = np.mean(biases[4]) + np.mean(biases[5])
mean_flat3 = np.mean(flats[4]) + np.mean(flats[5])

std_bias3 = np.std(biasdif3)
std_flat3 = np.std(flatdif3)

gain3 = (mean_flat3 - mean_bias3) / (std_flat3**2 - std_bias3**2)
readnoise3 = gain3 * (std_bias3) / (np.sqrt(2))

print(f'gain = {gain3}\nreadnoise = {readnoise3}')
```

```
gain = 0.26997617880249297
readnoise = 4.467316109555829
```

[353]:
```
biasdif4 = biases[6] - biases[7]
flatdif4 = flats[6] - flats[7]

mean_bias4 = np.mean(biases[6]) + np.mean(biases[7])
mean_flat4 = np.mean(flats[6]) + np.mean(flats[7])

std_bias4 = np.std(biasdif4)
std_flat4 = np.std(flatdif4)

gain4 = (mean_flat4 - mean_bias4) / (std_flat4**2 - std_bias4**2)
readnoise4 = gain4 * (std_bias4) / (np.sqrt(2))

print(f'gain = {gain4}\nreadnoise = {readnoise4}')
```

```
gain = 0.2703136343422885
readnoise = 4.479188725747097
```

[354]:
```
biasdif5 = biases[8] - biases[9]
flatdif5 = flats[8] - flats[9]

mean_bias5 = np.mean(biases[8]) + np.mean(biases[9])
mean_flat5 = np.mean(flats[8]) + np.mean(flats[9])

std_bias5 = np.std(biasdif5)
std_flat5 = np.std(flatdif5)
```

```
gain5 = (mean_flat5 - mean_bias5) / (std_flat5**2 - std_bias5**2)
readnoise5 = gain5 * (std_bias5) / (np.sqrt(2))

print(f'gain = {gain5}\nreadnoise = {readnoise5}')
```

```
gain = 0.27018172333707086
readnoise = 4.472157177618272
```

3.2 ALL readnoise/gain

[355]:
```
all_gains = []
all_readnoises = []

for i in range(0, len(biases), 2):
    bias_dif = biases[i] - biases[i+1]
    flat_dif = flats[i] - flats[i+1]

    mean_bias = np.mean(biases[i]) + np.mean(biases[i+1])
    mean_flat = np.mean(flats[i]) + np.mean(flats[i+1])

    std_bias = np.std(bias_dif)
    std_flat = np.std(flat_dif)

    gain = (mean_flat - mean_bias) / (std_flat**2 - std_bias**2)
    readnoise = gain * std_bias / np.sqrt(2)

    all_gains.append(gain)
    all_readnoises.append(readnoise)

    print(f'Pair {i//2 + 1}:\ngain = {gain}\nreadnoise = {readnoise}')

avg_gain = np.mean(all_gains)
avg_readnoise = np.mean(all_readnoises)
print(f"\nAverage Gain: {avg_gain:.4f}")
print(f"Average Readnoise: {avg_readnoise:.4f}")
```

```
Pair 1:
gain = 0.2696555886000731
readnoise = 4.470695042713425
Pair 2:
gain = 0.26984208637440976
readnoise = 4.471747003084644
Pair 3:
gain = 0.26997617880249297
readnoise = 4.467316109555829
Pair 4:
gain = 0.2703136343422885
readnoise = 4.479188725747097
```

```
Pair 5:
gain = 0.27018172333707086
readnoise = 4.472157177618272

Average Gain: 0.2700
Average Readnoise: 4.4722
```

4 plotting loop for lamps

[356]:
```
def plot_image_grid(file_list, title_prefix="Image"):
    fig, axes = plt.subplots(2, 5, figsize=(20, 8))
    axes = axes.flatten()  # single index for iteration

    for i, file in enumerate(file_list):
        if i >= 10: break # Stop if there are more than 10 files

        data = fits.getdata(file)

        im = axes[i].imshow(data, vmin=np.percentile(data, 1), vmax=np.
 ↪percentile(data, 99))
        axes[i].set_title(f"{title_prefix} {i}: {file}")
        axes[i].axis('off')
        axes[i].grid(False) # this and axis removed for cleaner look

    plt.tight_layout()
    plt.show()
```

5 Mercury

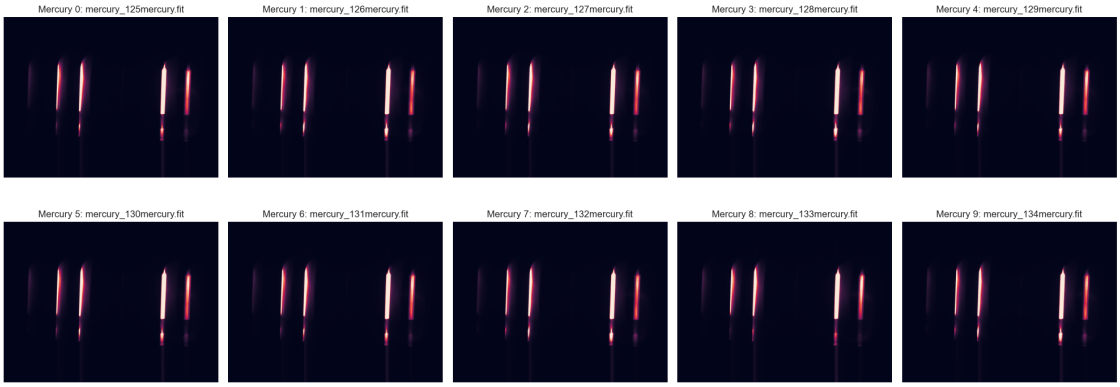
[357]:
```
imagem = fits.open('mercury_132mercury.fit')
imagem[0].data
print(np.min(imagem[0].data), np.max(imagem[0].data))
```

```
0 65535
```

[358]:
```
mercury = sorted(glob('mercury_*'))
print(mercury)
```

```
['mercury_125mercury.fit', 'mercury_126mercury.fit', 'mercury_127mercury.fit',
'mercury_128mercury.fit', 'mercury_129mercury.fit', 'mercury_130mercury.fit',
'mercury_131mercury.fit', 'mercury_132mercury.fit', 'mercury_133mercury.fit',
'mercury_134mercury.fit']
```

[359]:
```
plot_image_grid(mercury, title_prefix="Mercury")
```

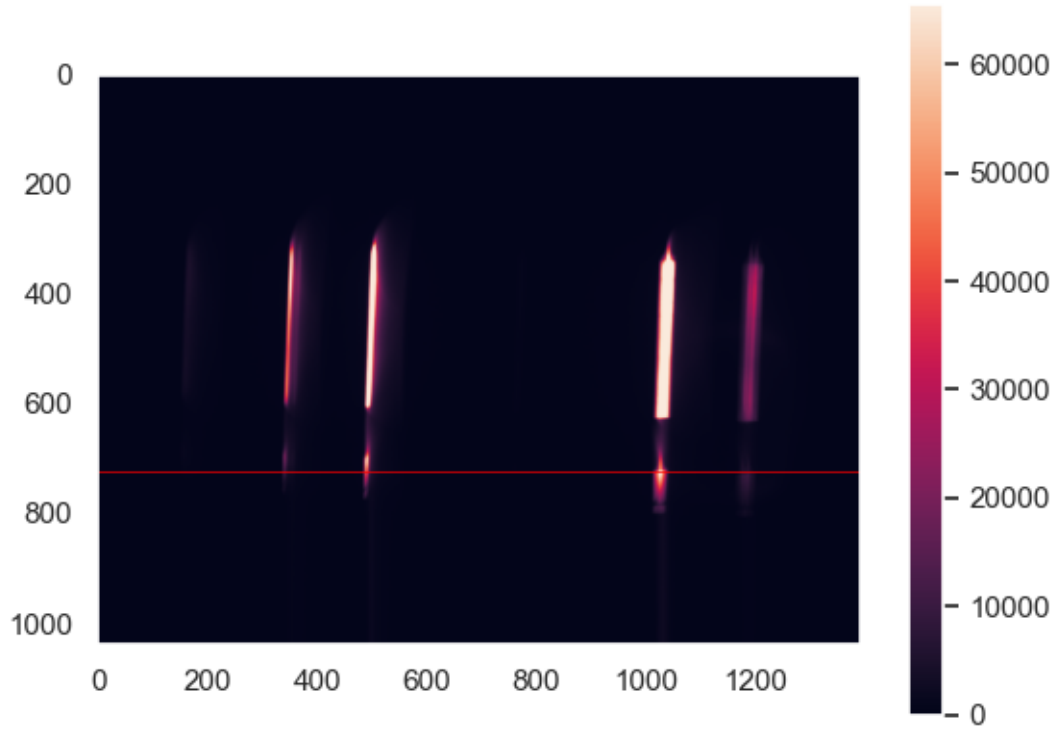


[360]:
```
for i in range(len(mercury)):
    print(f"Image {i}\nmin: {np.min(fits.getdata(mercury[i]))}, max: {np.
 ↪max(fits.getdata(mercury[i]))}")
```

```
Image 0
min: 217, max: 65535
Image 1
min: 215, max: 65535
Image 2
min: 214, max: 65535
Image 3
min: 215, max: 65535
Image 4
min: 216, max: 65535
Image 5
min: 208, max: 65535
Image 6
min: 217, max: 65535
Image 7
min: 0, max: 65535
Image 8
min: 218, max: 65535
Image 9
min: 211, max: 65535
```

[361]:
```
# Chose image 7 as the best exposure

imagem = fits.open('mercury_132mercury.fit')
merc = imagem[0].data

plt.imshow(merc, vmin=0, vmax=65535)
plt.axhline(y=725, color='red', linewidth=0.5)
plt.colorbar()
```

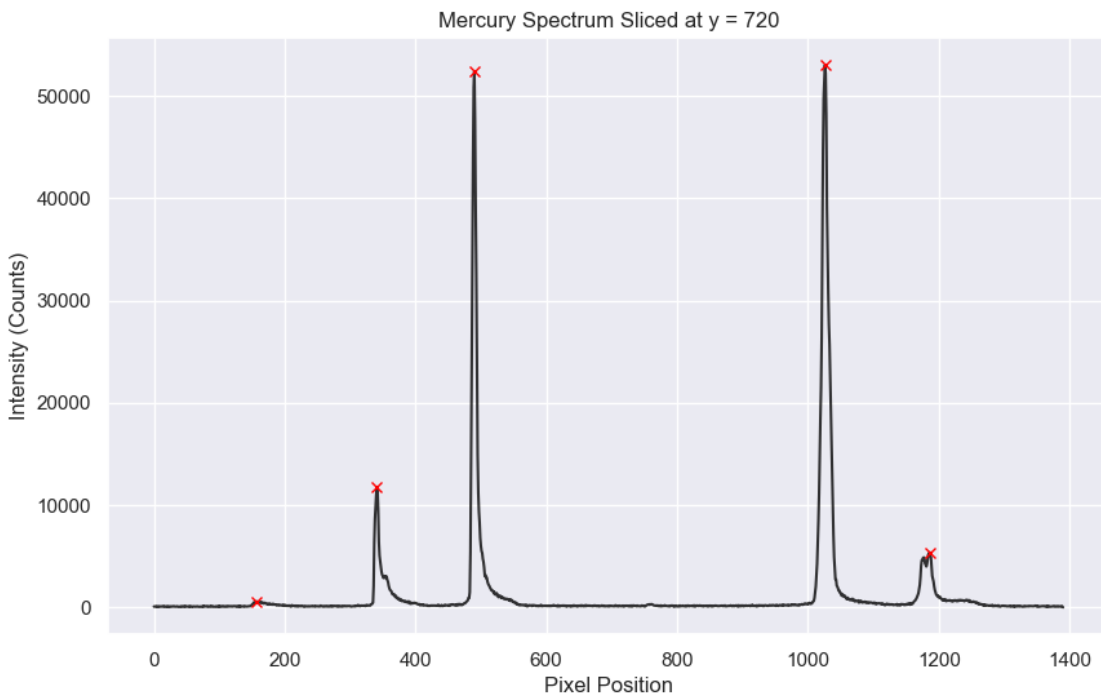
```
plt.grid(False)

plt.show()

print(merc.shape)
```




```
(1039, 1391)
```

[362]:
```
clean_merc = merc - master_bias

spec_slice_m = clean_merc[720, :]
peaks, properties = find_peaks(spec_slice_m, prominence=500, distance=10)

plt.figure(figsize=(10, 6))
plt.plot(spec_slice_m, color='black', alpha=0.8)
plt.plot(peaks, spec_slice_m[peaks], "x", color='red')

plt.title("Mercury Spectrum Sliced at y = 720")
plt.xlabel("Pixel Position")
plt.ylabel("Intensity (Counts)")

plt.show()
```

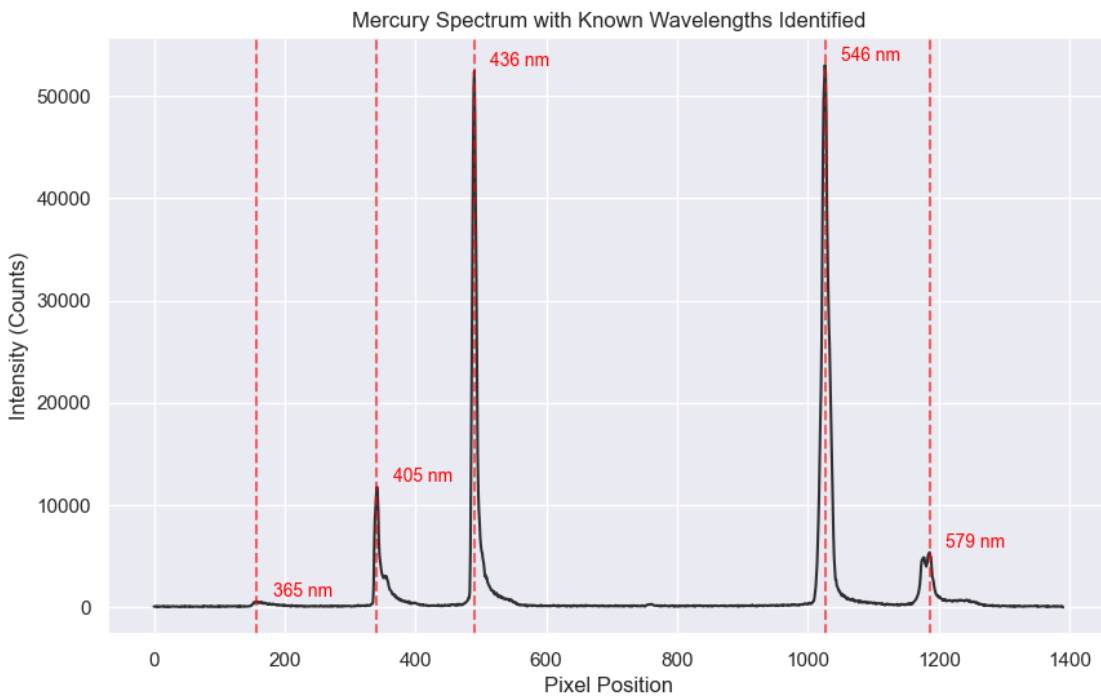
```
print(f"Peak X-positions (pixels): {peaks}")
for i, p in enumerate(peaks):
    print(f"Peak {i+1}: x = {p}, Intensity = {spec_slice_m[p]:.1f}")
```




```
Peak X-positions (pixels): [ 157  341  490 1026 1186]
Peak 1: x = 157, Intensity = 577.2
Peak 2: x = 341, Intensity = 11767.5
Peak 3: x = 490, Intensity = 52428.5
Peak 4: x = 1026, Intensity = 53010.9
Peak 5: x = 1186, Intensity = 5382.2
```

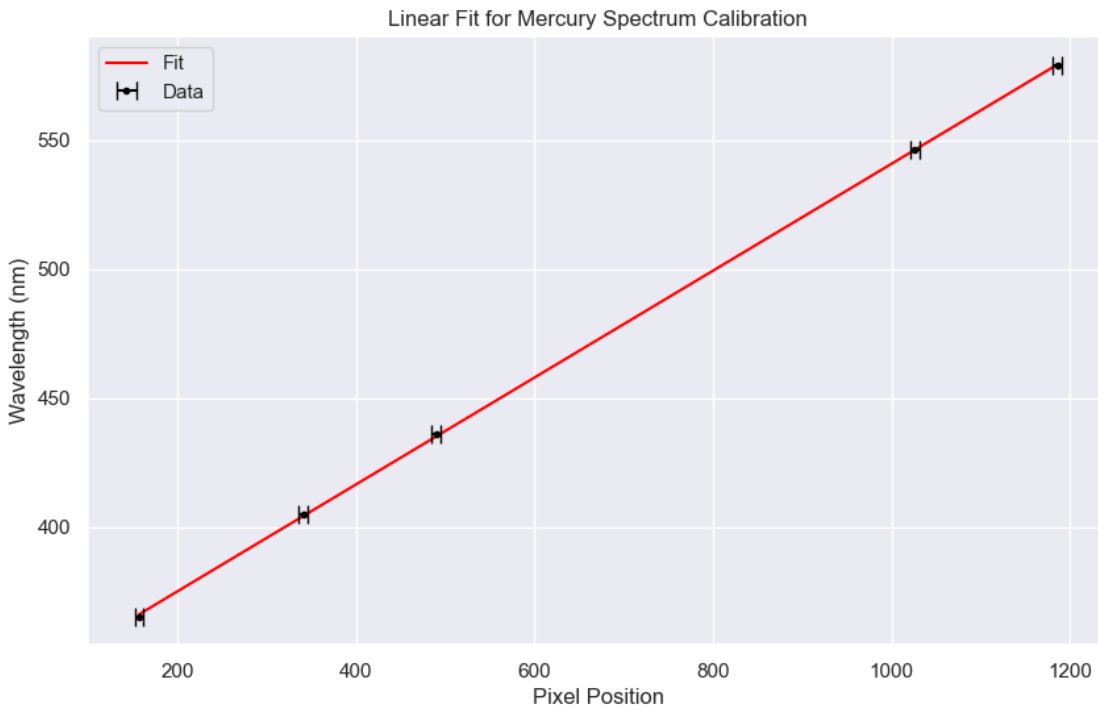
[363]:
```
pixel_peaks = [157, 341, 490, 1026, 1186]
wavelength_labels = ["365 nm", "405 nm", "436 nm", "546 nm", "579 nm"]

plt.figure(figsize=(10, 6))
plt.plot(spec_slice_m, color='black', alpha=0.8)

for px, label in zip(pixel_peaks, wavelength_labels):
    plt.axvline(x=px, color='red', linestyle='--', alpha=0.6)
    plt.text(px + 70, spec_slice_m[px] + 100, label,
             verticalalignment='bottom', horizontalalignment='center',␣
 ↪color='red', size=10)

plt.title("Mercury Spectrum with Known Wavelengths Identified")
```

```
plt.xlabel("Pixel Position")
plt.ylabel("Intensity (Counts)")
plt.show()
```



[390]:
```
x = np.array([157, 341, 490, 1026, 1186])
y = np.array([365, 405, 436, 546, 579])

merc_fit = np.polyfit(x, y, 1)
m, c = merc_fit

y_merc_fit = m * x + c
err = y - y_merc_fit
merc_err = np.std(err)

plt.figure(figsize=(10, 6))
plt.errorbar(x, y, xerr=5, fmt='.', capsize=5, label='Data', color='black',␣
↪zorder=2)
plt.plot(x, y_merc_fit, label='Fit', color='red', zorder=1)

plt.ylabel('Wavelength (nm)')
plt.xlabel('Pixel Position')
plt.title('Linear Fit for Mercury Spectrum Calibration')
plt.legend()
```

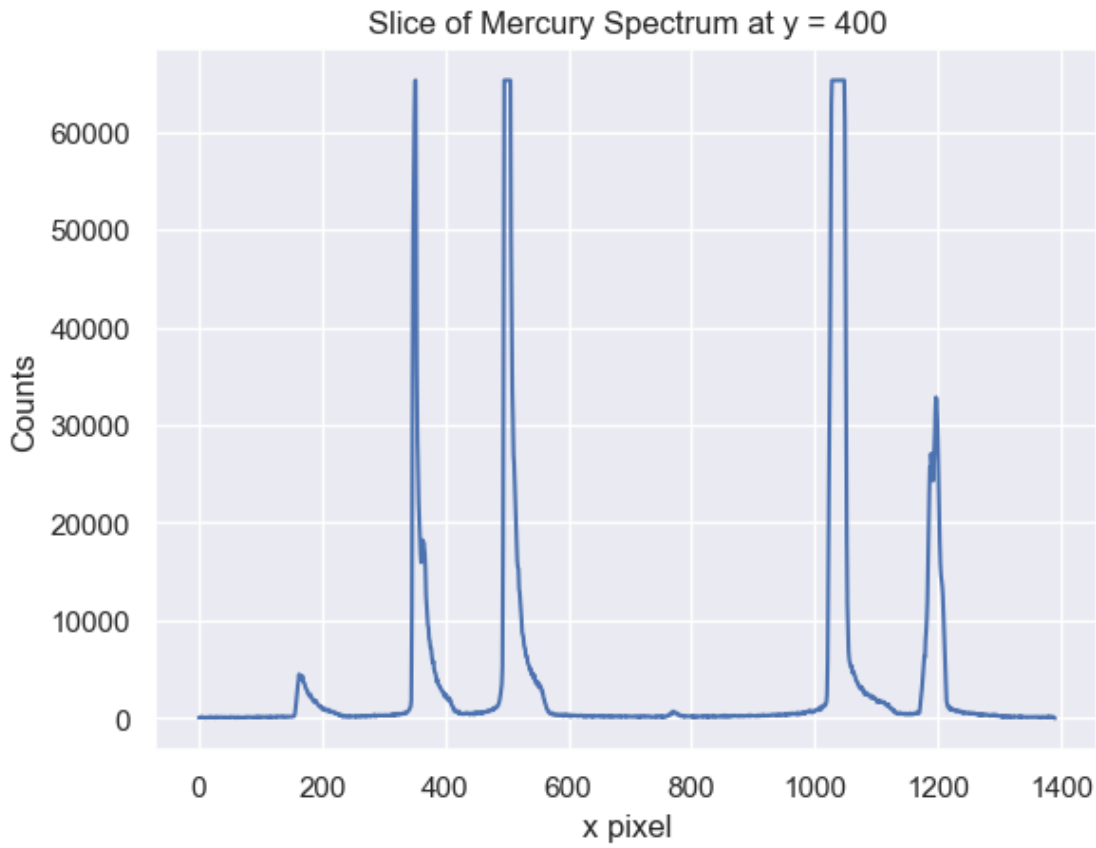
```
plt.show()

print(f"Slope = {m:.4f}, Intercept = {c:.4f}")
print(f"Mercury Calibration Margin of Error: {merc_err:.4f} nm")
```




```
Slope = 0.2071, Intercept = 333.6445
Mercury Calibration Margin of Error: 0.7399 nm
```

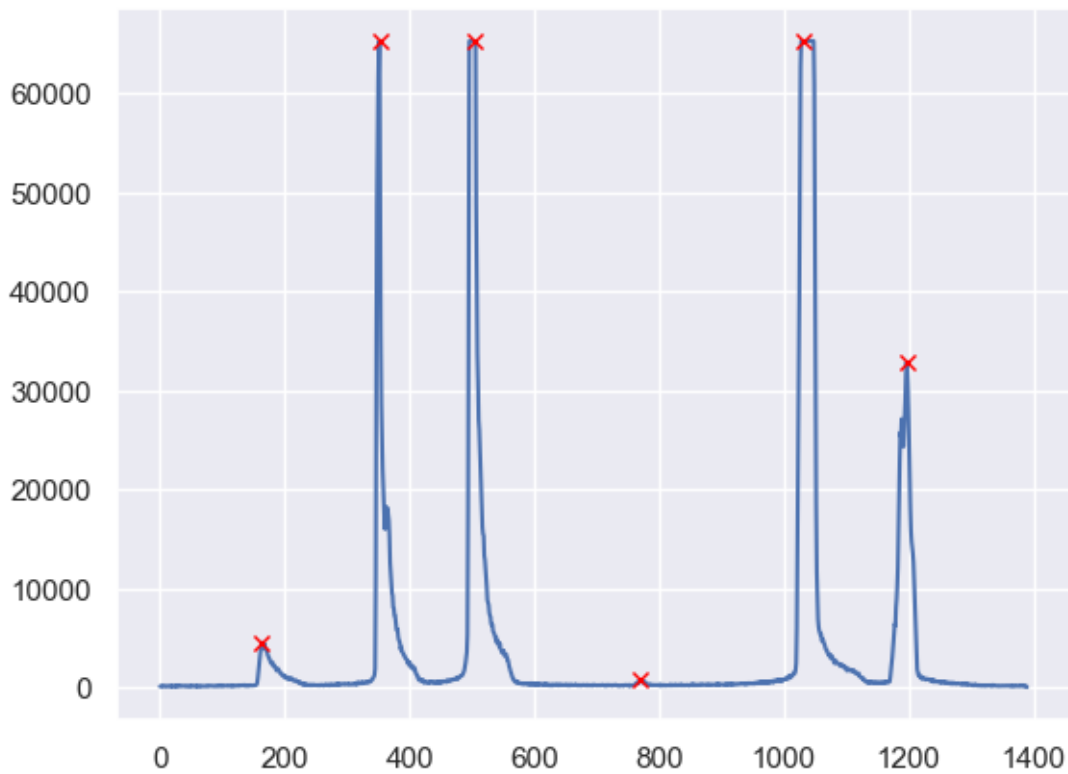
5.0.1 Saturated Slice (for discussion)

[365]:
```
plt.plot(clean_merc[400, :])
plt.title("Slice of Mercury Spectrum at y = 400")

plt.xlabel("x pixel")
plt.ylabel("Counts")

plt.show()
```



[366]:
```
spec_slice = clean_merc[400, :]
peaks, properties = find_peaks(spec_slice, prominence=500, distance=15)

plt.plot(spec_slice)
plt.plot(peaks, spec_slice[peaks], "x", color='red')

# 4. Print the X-positions (pixel indices)
print(f"Peak X-positions (pixels): {peaks}")
for i, p in enumerate(peaks):
    print(f"Peak {i+1}: x = {p}, Intensity = {spec_slice[p]:.1f}")

plt.show()
```

```
Peak X-positions (pixels): [ 163  352  503  769 1032 1197]
Peak 1: x = 163, Intensity = 4503.6
Peak 2: x = 352, Intensity = 65268.9
Peak 3: x = 503, Intensity = 65276.5
Peak 4: x = 769, Intensity = 693.2
Peak 5: x = 1032, Intensity = 65281.2
Peak 6: x = 1197, Intensity = 32836.4
```

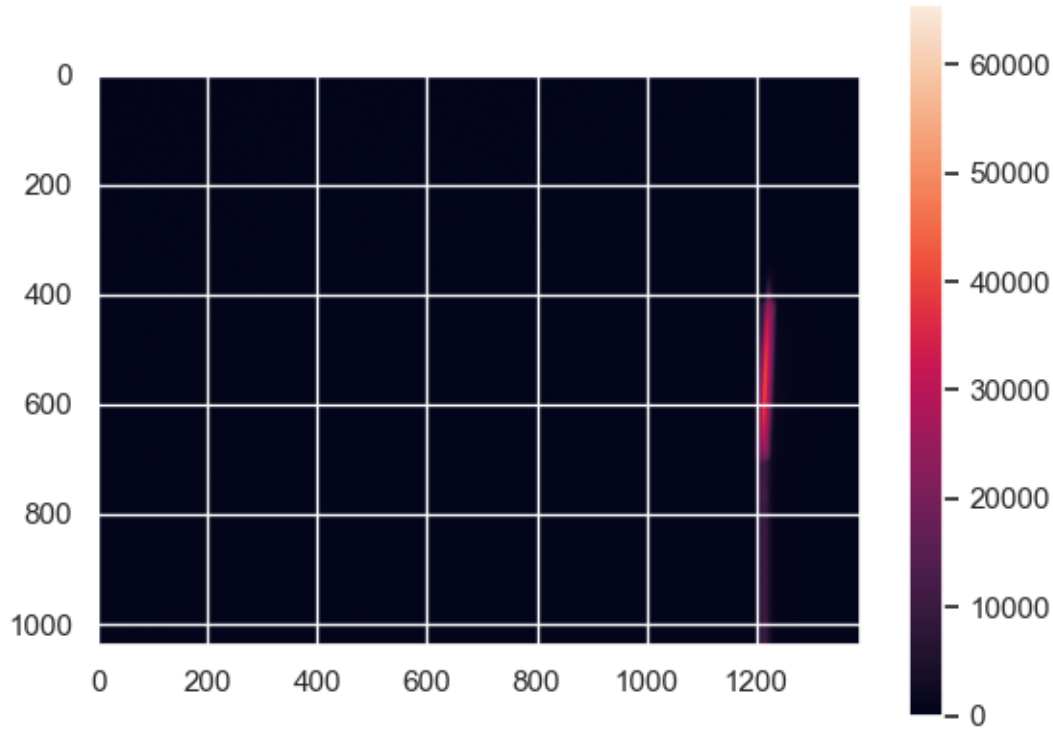


6 Sodium

[367]:
```
images = fits.open('sodium_009sodium.fit')
images[0].data
print(np.min(images[0].data), np.max(images[0].data))

sod = images[0].data
plt.imshow(sod, vmin=0 ,vmax=65535)
plt.colorbar()
```

```
189 42236
```

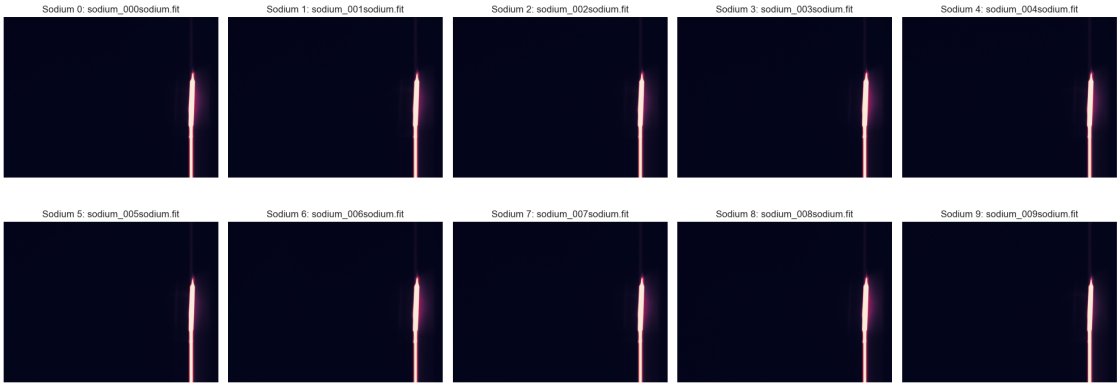
[367]: <matplotlib.colorbar.Colorbar at 0x11cca08d0>



[368]:
```
sodium = sorted(glob('*sodium.fit'))
print(sodium)
```

```
['sodium_000sodium.fit', 'sodium_001sodium.fit', 'sodium_002sodium.fit',
'sodium_003sodium.fit', 'sodium_004sodium.fit', 'sodium_005sodium.fit',
'sodium_006sodium.fit', 'sodium_007sodium.fit', 'sodium_008sodium.fit',
'sodium_009sodium.fit']
```

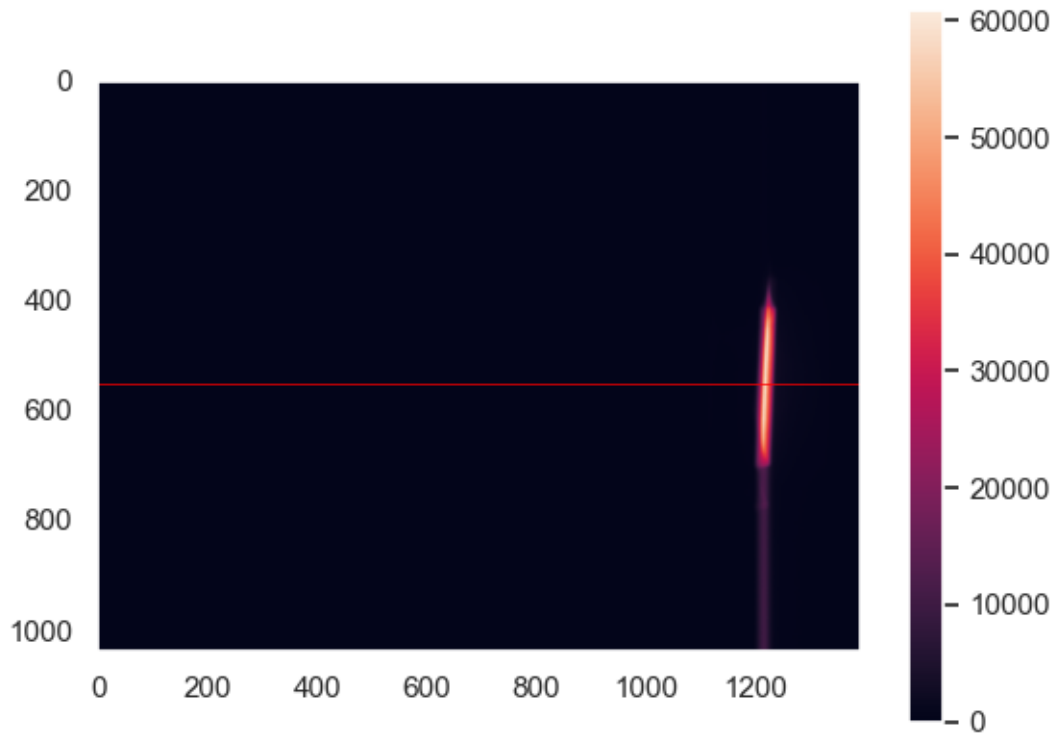
[369]:
```
plot_image_grid(sodium, title_prefix="Sodium")
```



[370]:
```
for i in range(len(sodium)):
    print(f"Image {i}\nmin: {np.min(fits.getdata(sodium[i]))}, max: {np.
 ↪max(fits.getdata(sodium[i]))}")
```

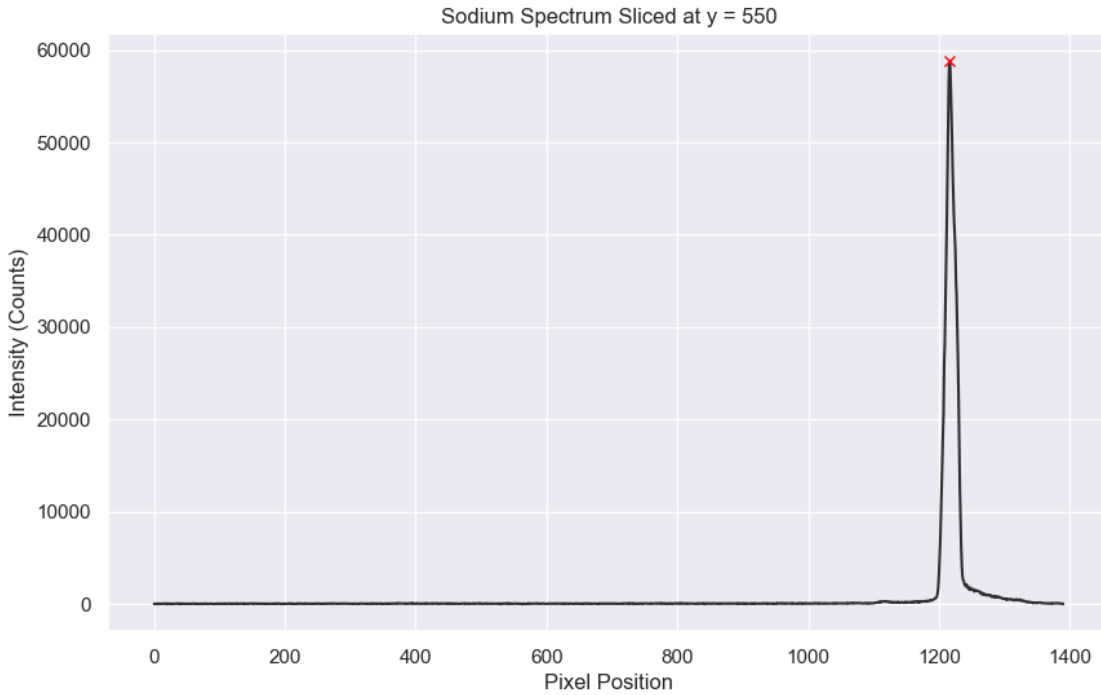
```
Image 0
min: 193, max: 65535
Image 1
min: 193, max: 60921
Image 2
min: 189, max: 59868
Image 3
min: 193, max: 65535
Image 4
min: 186, max: 65535
Image 5
min: 190, max: 49077
Image 6
min: 201, max: 65535
Image 7
min: 191, max: 65535
Image 8
min: 187, max: 65535
Image 9
min: 189, max: 42236
```

[371]:
```
# Chose image 1 as the best exposure

images = fits.open('sodium_001sodium.fit')
sod = images[0].data

plt.imshow(sod, vmin=0, vmax=60921)
plt.axhline(y=550, color='red', linewidth=0.5)
plt.colorbar()
plt.grid(False)

plt.show()

print(sod.shape)
```




```
(1039, 1391)
```

[372]:
```python
clean_sod = sod - master_bias

spec_slice_s = clean_sod[550, :]
peaks, properties = find_peaks(spec_slice_s, prominence=500, distance=10)

plt.figure(figsize=(10, 6))
plt.plot(spec_slice_s, color='black', alpha=0.8)
plt.plot(peaks, spec_slice_s[peaks], "x", color='red')

plt.title("Sodium Spectrum Sliced at y = 550")
plt.xlabel("Pixel Position")
plt.ylabel("Intensity (Counts)")

plt.show()

print(f"Peak X-positions (pixels): {peaks}")
for i, p in enumerate(peaks):
    print(f"Peak {i+1}: x = {p}, Intensity = {spec_slice_s[p]:.1f}")
```




```
Peak X-positions (pixels): [1216]
Peak 1: x = 1216, Intensity = 58800.4
```

[373]:
```python
m = 0.2071
c = 333.6445

def wavelenght_cali(px):
    return m * px + c

print(wavelenght_cali(1216))

# known sodium D-line at 589.3 nm
deviation = np.abs(wavelenght_cali(1216) - 589.3)
print(f'Sodium Deviation: {deviation} nm')

if deviation <= merc_err:
    print("The sodium measurement is within the margin of error of the mercury␣
 ↪calibration.")
else:
    print("The sodium measurement is outside the margin of error.")
```

```
585.4781
Sodium Deviation: 3.821899999999914 nm
The sodium measurement is outside the margin of error.
```

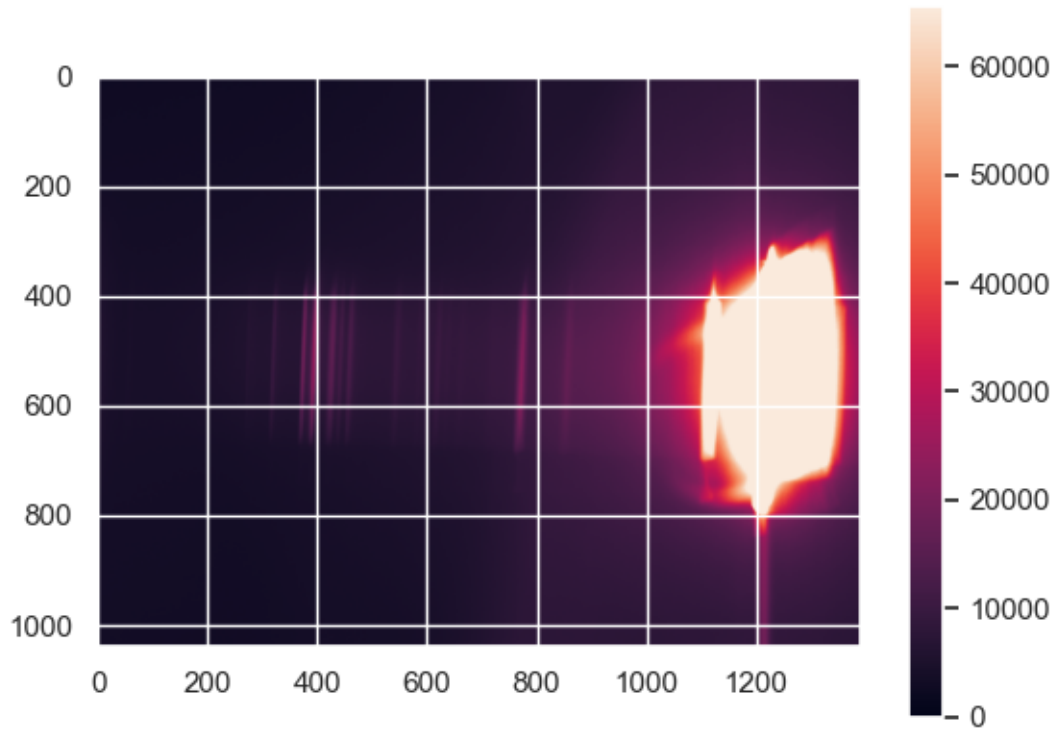
6.0.1 Sodium: Saturated (for discussion)

[374]:
```
imagess = fits.open('sodium_004sodium-saturated.fit')
imagess[0].data
print(np.min(imagess[0].data), np.max(imagess[0].data))
```

```
222 65535
```

[375]:
```
sodsat = imagess[0].data
plt.imshow(sodsat, vmin=0 ,vmax=65535)
plt.colorbar()
```

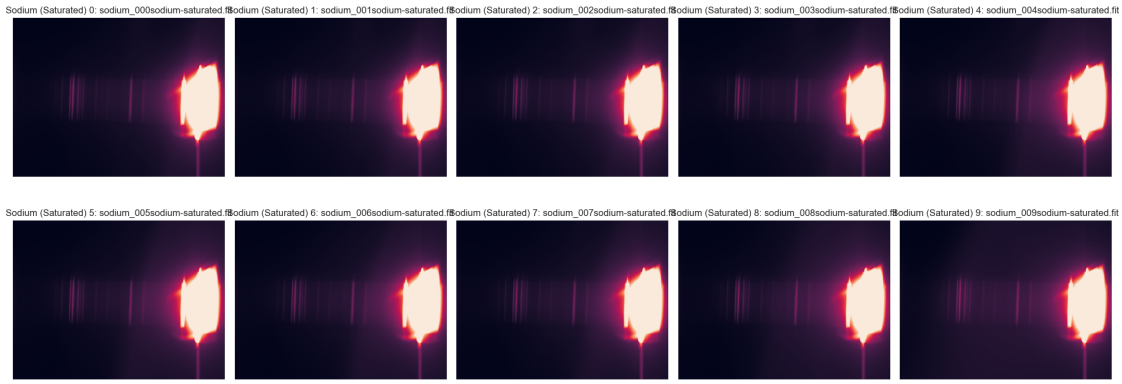
[375]:
```
<matplotlib.colorbar.Colorbar at 0x11d2f49d0>
```



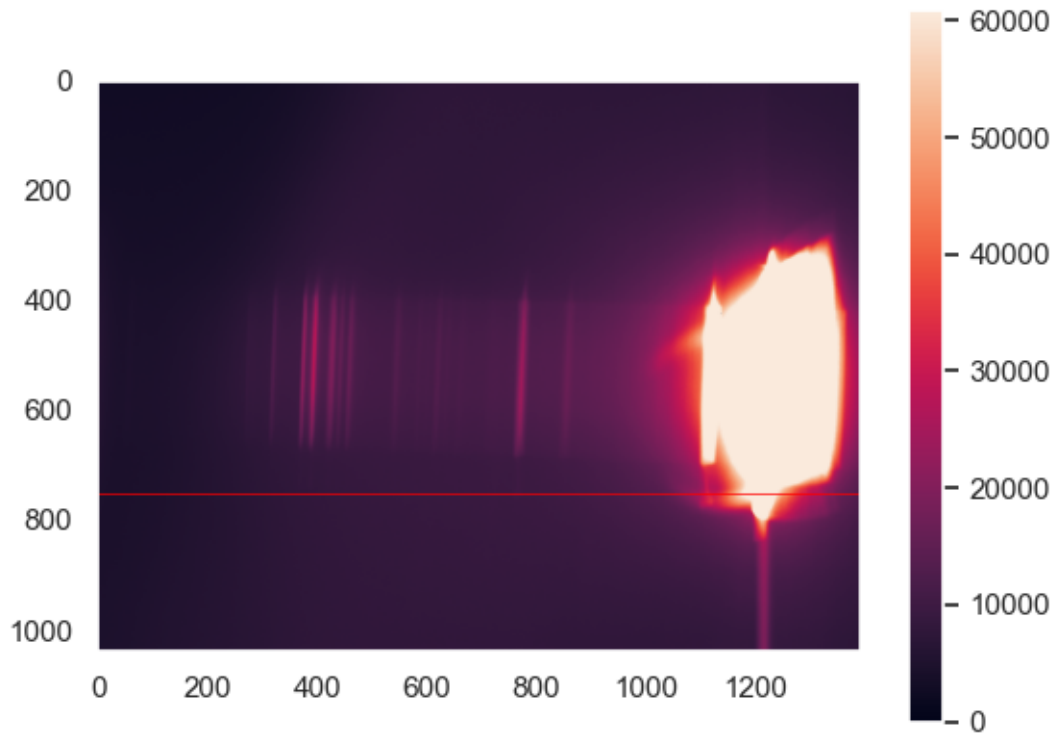
[376]:
```
sodium_sat = sorted(glob('*sodium-saturated*'))
print(sodium_sat)
```

```
['sodium_000sodium-saturated.fit', 'sodium_001sodium-saturated.fit',
'sodium_002sodium-saturated.fit', 'sodium_003sodium-saturated.fit',
'sodium_004sodium-saturated.fit', 'sodium_005sodium-saturated.fit',
'sodium_006sodium-saturated.fit', 'sodium_007sodium-saturated.fit',
'sodium_008sodium-saturated.fit', 'sodium_009sodium-saturated.fit']
```

[377]:
```
plot_image_grid(sodium_sat, title_prefix="Sodium (Saturated)")
```

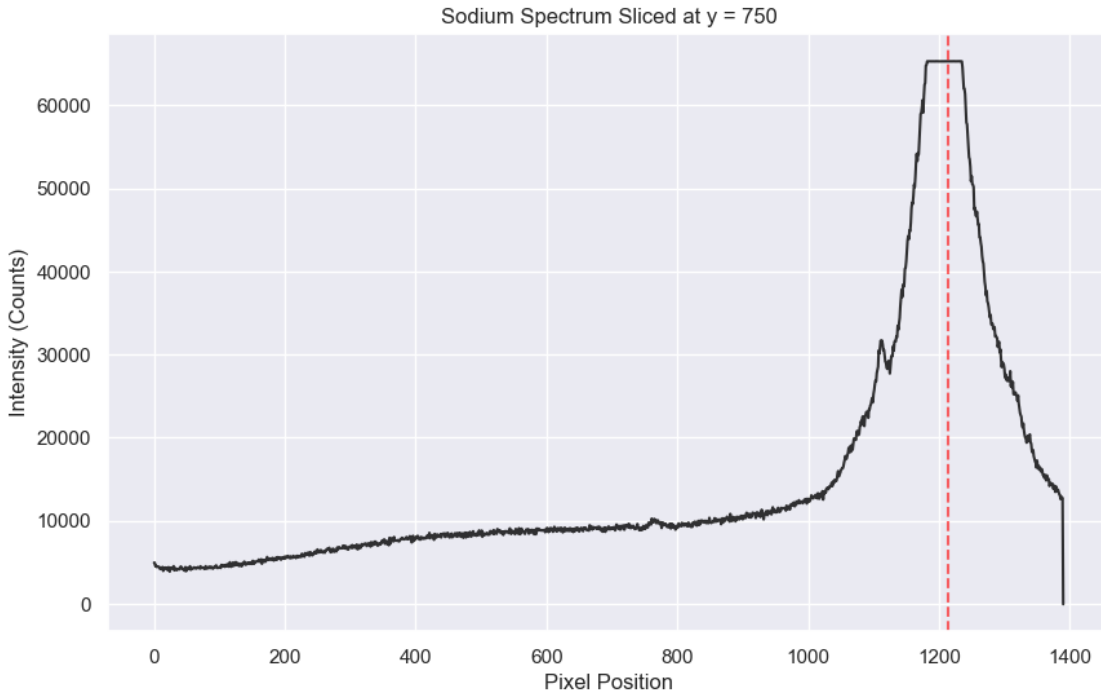


[378]:
```
# Chose image 9 as the best exposure

imagess = fits.open('sodium_009sodium-saturated.fit')
sodsat = imagess[0].data

plt.imshow(sodsat, vmin=0, vmax=60921)
plt.axhline(y=750, color='red', linewidth=0.5)
plt.colorbar()
plt.grid(False)

plt.show()

print(sodsat.shape)
```

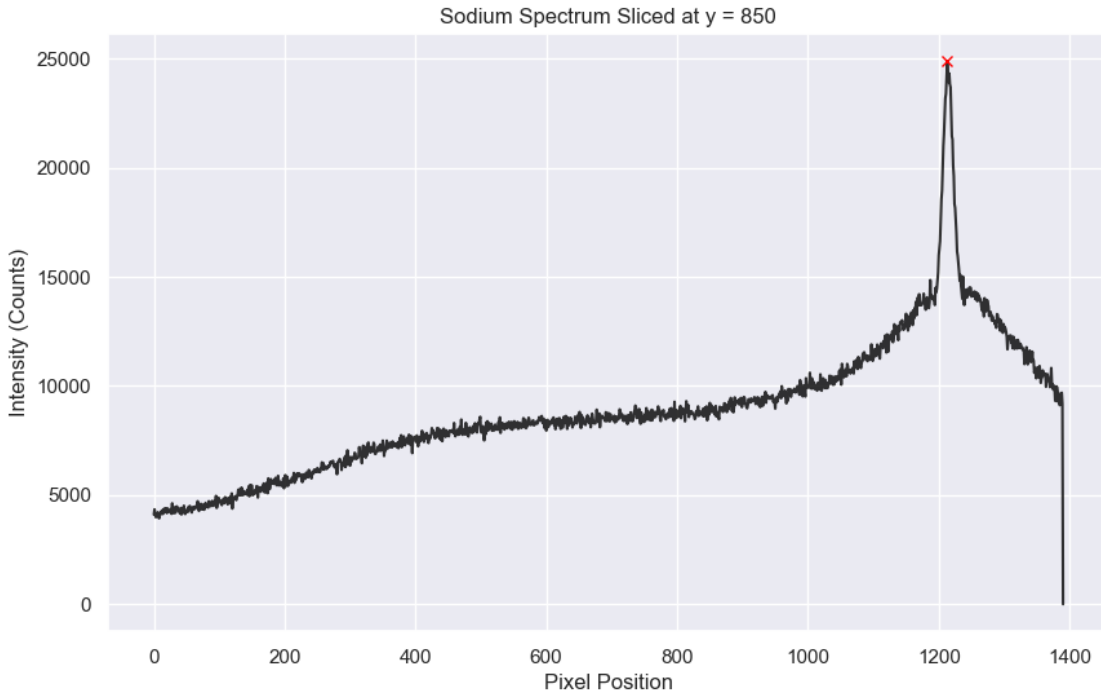



```
(1039, 1391)
```

[379]:
```
imagess = fits.open('sodium_009sodium-saturated.fit')
sodsat = imagess[0].data

clean_sod_sat = sodsat - master_bias

plt.figure(figsize=(10, 6))
plt.plot(clean_sod_sat[750, :], color='black', alpha=0.8)
plt.axvline(x=1213, color='red', linestyle='--', alpha=0.6)

plt.title("Sodium Spectrum Sliced at y = 750")
plt.xlabel("Pixel Position")
plt.ylabel("Intensity (Counts)")

plt.show()
```



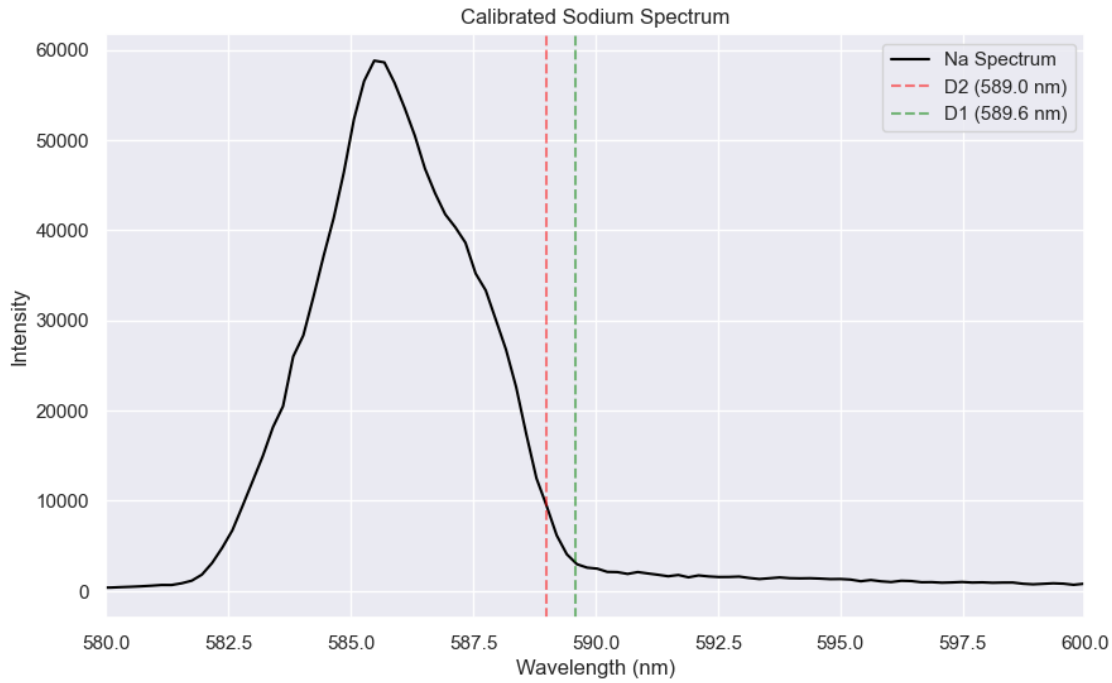
[380]:
```
spec_slice_ss = clean_sod_sat[850, :]
peaks, properties = find_peaks(spec_slice_ss, prominence=1500, distance=10)

plt.figure(figsize=(10, 6))
plt.plot(spec_slice_ss, color='black', alpha=0.8)
plt.plot(peaks, spec_slice_ss[peaks], "x", color='red')

plt.title("Sodium Spectrum Sliced at y = 850")
plt.xlabel("Pixel Position")
plt.ylabel("Intensity (Counts)")

plt.show()

print(f"Peak X-positions (pixels): {peaks}")
for i, p in enumerate(peaks):
    print(f"Peak {i+1}: x = {p}, Intensity = {spec_slice_ss[p]:.1f}")
```




```
Peak X-positions (pixels): [1213]
Peak 1: x = 1213, Intensity = 24895.2
```

7 Resolution (Gaussian)

[381]:
```
wavelengths = m * np.arange(len(spec_slice_s)) + c

plt.figure(figsize=(10, 6))
plt.plot(wavelengths, spec_slice_s, color='black', label='Na Spectrum')

plt.axvline(x=589.0, color='red', linestyle='--', alpha=0.5, label='D2 (589.0␣
↪nm)')
plt.axvline(x=589.6, color='green', linestyle='--', alpha=0.5, label='D1 (589.6␣
↪nm)')

plt.title("Calibrated Sodium Spectrum")
plt.xlabel("Wavelength (nm)")
plt.ylabel("Intensity")
plt.xlim(580, 600)
plt.legend()

plt.show()
```



[382]:
```
def gaussian(x, a, x0, sigma, offset):
    return a * np.exp(-(x - x0)**2 / (2 * sigma**2)) + offset

# For the 546 nm peak (Pixel 1026)
window = 15 # pixels around the peak
x_data = wavelengths[1026 - window : 1026 + window]
y_data = spec_slice_m[1026 - window : 1026 + window]

# initial guess for amplitude, mean, sigma, offset
popt, pcov = curve_fit(gaussian, x_data, y_data, p0=[max(y_data), 546, 1,␣
 ↪min(y_data)])
errors = np.sqrt(np.diag(pcov))

err_amplitude = errors[0] # popt[0]: amplitude
err_center = errors[1] # popt[1]: centre
err_sigma = errors[2] # popt[2]: sigma/width
err_offset = errors[3] # popt[3]: offset

print(f"Fitted Center: {popt[1]:.4f} ± {err_center:.4f} nm")
print(f"Fitted Width: {popt[2]:.4f} ± {err_sigma:.4f} nm")

sigma = popt[2]
fwhm = 2.355 * abs(sigma)  # convert sigma to FWHM
resolution = 546.0 / fwhm
```

```
err_fwhm = 2.355 * err_sigma
err_R = resolution * np.sqrt((err_fwhm / fwhm)**2 + (err_center / popt[1])**2)

print(f"FWHM (Δ ): {fwhm:.3f} ± {err_fwhm:.3f} nm")
print(f"Spectral Resolution (R): {resolution:.1f} ± {err_R:.1f}")
```

```
Fitted Center: 546.2202 ± 0.0339 nm
Fitted Width: 1.0489 ± 0.0571 nm
FWHM (Δ ): 2.470 ± 0.135 nm
Spectral Resolution (R): 221.0 ± 12.0
```

[383]:
```
# For the 436 nm peak (Pixel 490)
window = 15 # pixels around the peak
x_data = wavelengths[490 - window : 490 + window]
y_data = spec_slice_m[490 - window : 490 + window]

# initial guess for amplitude, mean, sigma, offset
popt, pcov = curve_fit(gaussian, x_data, y_data, p0=[max(y_data), 436, 1,␣
 ↪min(y_data)])
errors = np.sqrt(np.diag(pcov))

err_amplitude = errors[0] # popt[0]: amplitude
err_center = errors[1] # popt[1]: centre
err_sigma = errors[2] # popt[2]: sigma/width
err_offset = errors[3] # popt[3]: offset

print(f"Fitted Center: {popt[1]:.4f} ± {err_center:.4f} nm")
print(f"Fitted Width: {popt[2]:.4f} ± {err_sigma:.4f} nm")

sigma = popt[2]
fwhm = 2.355 * abs(sigma)  # convert sigma to FWHM
resolution = 436.0 / fwhm

err_fwhm = 2.355 * err_sigma
err_R = resolution * np.sqrt((err_fwhm / fwhm)**2 + (err_center / popt[1])**2)

print(f"FWHM (Δ ): {fwhm:.3f} ± {err_fwhm:.3f} nm")
print(f"Spectral Resolution (R): {resolution:.1f} ± {err_R:.1f}")
```

```
Fitted Center: 435.1193 ± 0.0204 nm
Fitted Width: 0.6254 ± 0.0240 nm
FWHM (Δ ): 1.473 ± 0.056 nm
Spectral Resolution (R): 296.0 ± 11.3
```

[384]:
```
# For the 405 nm peak (Pixel 341)
window = 15 # pixels around the peak
x_data = wavelengths[341 - window : 341 + window]
```

```
y_data = spec_slice_m[341 - window : 341 + window]

# initial guess for amplitude, mean, sigma, offset
popt, pcov = curve_fit(gaussian, x_data, y_data, p0=[max(y_data), 405, 1,␣
 ↪min(y_data)])
errors = np.sqrt(np.diag(pcov))

err_amplitude = errors[0] # popt[0]: amplitude
err_center = errors[1] # popt[1]: centre
err_sigma = errors[2] # popt[2]: sigma/width
err_offset = errors[3] # popt[3]: offset

print(f"Fitted Center: {popt[1]:.4f} ± {err_center:.4f} nm")
print(f"Fitted Width: {popt[2]:.4f} ± {err_sigma:.4f} nm")

sigma = popt[2]
fwhm = 2.355 * abs(sigma)  # convert sigma to FWHM
resolution = 405.0 / fwhm

err_fwhm = 2.355 * err_sigma
err_R = resolution * np.sqrt((err_fwhm / fwhm)**2 + (err_center / popt[1])**2)

print(f"FWHM (Δ ): {fwhm:.3f} ± {err_fwhm:.3f} nm")
print(f"Spectral Resolution (R): {resolution:.1f} ± {err_R:.1f}")
```

```
Fitted Center: 404.2695 ± 0.0486 nm
Fitted Width: 0.5909 ± 0.0562 nm
FWHM (Δ ): 1.391 ± 0.132 nm
Spectral Resolution (R): 291.1 ± 27.7
```

[385]:
```
# For the 579 nm peak (Pixel 1186)
window = 15 # pixels around the peak
x_data = wavelengths[1186 - window : 1186 + window]
y_data = spec_slice_m[1186 - window : 1186 + window]

# initial guess for amplitude, mean, sigma, offset
popt, pcov = curve_fit(gaussian, x_data, y_data, p0=[max(y_data), 579, 1,␣
 ↪min(y_data)])
errors = np.sqrt(np.diag(pcov))

err_amplitude = errors[0] # popt[0]: amplitude
err_center = errors[1] # popt[1]: centre
err_sigma = errors[2] # popt[2]: sigma/width
err_offset = errors[3] # popt[3]: offset

print(f"Fitted Center: {popt[1]:.4f} ± {err_center:.4f} nm")
print(f"Fitted Width: {popt[2]:.4f} ± {err_sigma:.4f} nm")
```

```python
sigma = popt[2]
fwhm = 2.355 * abs(sigma)  # convert sigma to FWHM
resolution = 579.0 / fwhm

err_fwhm = 2.355 * err_sigma
err_R = resolution * np.sqrt((err_fwhm / fwhm)**2 + (err_center / popt[1])**2)

print(f"FWHM (Δ ): {fwhm:.3f} ± {err_fwhm:.3f} nm")
print(f"Spectral Resolution (R): {resolution:.1f} ± {err_R:.1f}")
```

```
Fitted Center: 578.2830 ± 0.1012 nm
Fitted Width: 2.1309 ± 0.4147 nm
FWHM (Δ ): 5.018 ± 0.977 nm
Spectral Resolution (R): 115.4 ± 22.5
```